

Frontier Inventory at the Close of Wave 12 (2026-04-22)

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Abstract

A structured distillation of the open research frontier surviving the wave-12 adversarial synthesis on the CY-to-chiral functor $\Phi : \text{CY-cat}_d \rightarrow \text{ChirAlg}$. Each item is stated as mathematics, scoped to its correct dimension and ambient, and pointed to the wave-12 note in which its detailed treatment survives. The inventory organises the frontier along five cross-sections: the CY-A/B/C/D/H framework, the universal Borchers weight identity $\kappa_{\text{BKM}}(\Phi_N) = c_N(0)/2$, structural conjectures, physics interweave, and cross-volume bridges; it closes with single-sentence targets for the next adversarial wave.

1 Frontier by CY-A/B/C/D/H framework

1.1 CY-A completion items

Chain-level $\text{Sp}_{K3,E}^{\text{ch}}$ beyond the existing proposition. The platonic-synthesis theorem furnishes $\text{Sp}_{K3,E}^{\text{ch}}(\Phi_3^{\text{FA}}(K3 \times E)) \simeq U^{\text{ch}}(\mathfrak{heis}_{\text{Muk}}) \otimes U^{\text{ch}}(\mathfrak{g}_{K3}^{\text{BPS}})$ as a stratum- S_3 derivation combining Costello–Li 6d hCS on the toric $\mathbb{C}^3$ -chart with Dolbeault gluing. The chain-level specialisation beyond a single reference curve — a factorisation-homology equation on $\text{Ran}(E)$ pulling back the K3-factor Maulik–Okounkov stable envelope to its E_1 -chiral shadow on E — has not been written. Obstruction: the E -factor E_1 -extraction on the rank-24 Mukai Heisenberg via Dunn–Lurie $\int_{K3} E_3 \simeq E_1$ needs the global critical-chart hypothesis, not automatic for every compact CY_3 . See `notes/wave12_f5_programme_meta_audit.tex` and `notes/wave12_u1_two_stage_functor.tex`.

S^3 -framing trivialisation on the quintic. The S^3 -framing obstruction for a CY_3 category lives in $\pi_3(B\text{Sp})$ and vanishes on every smooth projective CY_3 for topological reasons; no explicit chain-level trivialisation of the framing on $\text{Perf}(Q^5)$ is known outside the toric setting. The question is: write the trivialising A_∞ -twisting datum on the derived endomorphism algebra of the structure sheaf of the quintic. See `notes/wave12_a3_cy_a3_equivalence.tex` (Open question: explicit chain-level framing on the quintic).

Inverse direction Ω_3 . Under what hypotheses on $\mathcal{H} \in \text{Alg}_{E_1}$ does there exist a CY_3 category $\mathcal{C}$ with $\Phi_3(\mathcal{C}) \simeq \mathcal{H}$? This is CY-B₃ in disguise: the cobar functor Ω_3 must land back in $\text{CY}_3\text{-cat}$. See `notes/wave12_a3_cy_a3_equivalence.tex`.

Essential-image characterisation of Φ_3 . Describe the essential image of Φ_3 inside Alg_{E_1} . The working hypothesis: those E_1 -algebras admitting a compatible factorisation structure on every curve, plus Kontsevich–Soibelman wall-crossing flatness along the Stokes-stratification of the stability-condition moduli. See `notes/wave12_a3_cy_a3_equivalence.tex`.

Morphism preservation on compact non-product $\mathbf{CY}_3$. The functor Φ_3 is established as an $(\infty, 1)$ -functor at the object level; full morphism-level coherence (action on Fourier–Mukai kernels with cycle-class lifts) is not stated in the non-toric, non-product setting. Six-route convergence to $G(K3 \times E)$ and the Super–Yangian bridge both assume this. See `notes/wave12_a3_cy_a3_equivalence.tex` and `notes/wave12_a12_six_routes_k3_e.tex`.

1.2 CY-B Koszul duality

B_d at $d = 3$ conditional on which many stratum-S2 conjectures upgrade. Bar–cobar Quillen equivalence at $d = 3$ is the Koszul-duality wall; its chiral instantiation would promote the bar–cobar adjunction $\Phi_3 \dashv \Omega_3$ to an equivalence and rigidify the cobar presentation $\mathcal{H}(\mathcal{C}) = \Omega_3(\text{coalg on } \mathcal{C})$. See `notes/wave12_a3_cy_a3_equivalence.tex`.

Booth–Lazarev curved Quillen equivalence at compact $\mathbf{CY}_d$ with $\kappa_{\text{ch}} \neq 0$. Booth–Lazarev (arXiv:2304.08409) supply the abstract coderived + curved A_∞ Quillen equivalence; the chiral instantiation on $\overline{M}_{g,n}$ — i.e. the curved A_∞ cotangent complex in the chiral lane — is the Vol I $\leftrightarrow$ Vol III concordance point and remains open. See `notes/wave12_f5_programme_meta_audit.tex` and `notes/wave12_b2_cross_volume_consistency.tex`.

Chain-level $(B_{E_1}, B_{E_2}, B_{E_\infty})$ filtered equivalence. The three-sector stratification of the bar complex by $E_1/E_2/E_\infty$ layers — Vol III’s reading of the Vol II two-coloured $\mathbf{SC}^{\text{ch}, \text{top}}$ dioperad — is stated as a decomposition but not as a chain-level equivalence of filtered bar complexes. See `notes/wave12_f5_programme_meta_audit.tex`.

1.3 CY-C braided quantum group

Explicit $G(K3 \times E)$ construction. The braided E_2 -quantum group $G(K3 \times E) = \mathcal{Z}(\text{Rep}^{E_1}(\mathcal{A}_{K3 \times E}))$ is written as a Drinfeld centre, not as an explicit generator-and-relation Hopf algebroid. The CY-C three-routes convergence $C(g, q) = D^b(Y^+(\mathfrak{g}_{K3}))$ via CoHA/Koszul/stable-envelope is conjectural. See `notes/wave12_a10_cy_c_three_routes_soibelman.tex`.

Six-route convergence at the E_2 -braided level. The six constructions of $G(K3 \times E)$ — Borchers lift, lattice VOA, Kummer, CoHA via Φ , Maulik–Okounkov stable envelope, BFN Coulomb, BLLPR Schur — are six distinct constructions (cache fact, not six Φ -applications). Whether any subset converges rigorously at the E_2 -braided level (beyond the matching $\kappa_\bullet$ -spectrum $\{2, 3, 5, 24\}$) is conjectural. See `notes/wave12_a12_six_routes_k3_e.tex` and `notes/wave12_a3_cy_a3_equivalence.tex`.

$\mathcal{Z}(\text{Rep}^{E_1}(A_X))$ computational realisation. At $d \geq 3$, A_X is E_1 -chiral; the E_2 -structure lives on $\mathcal{Z}(\text{Rep}^{E_1}(A_X))$, not on A_X . A direct-computation realisation (generators, braiding matrices, universal R -matrix) on A_{K3} or $A_{K3 \times E}$ has not been exhibited. See `notes/wave12_f1_6d_hcs_e3_generators.tex` (item Q3).

Super–Yangian $Y(\mathfrak{gl}(4|20))$ explicit coproduct and R -matrix. The super-rank $(4, 20)$ split is the Mukai Hodge involution $\pm$ -eigendecomposition of $\tilde{\Lambda}_{K3}$; the embedding $Y(\mathfrak{gl}(4|20)) \hookrightarrow \mathbf{D}_h(\mathfrak{g}_{\Delta_5})$ is conjectural at the Hopf-superalgebra level. The explicit coproduct on the super-monodromy matrix $T(u)$ and the universal R -matrix (degree $(24, 24)$ in the geometric Maulik–Okounkov reading, $(16, 16)$ in the intrinsic Cartan-matrix reading) await construction. See `notes/wave12_a15_e8xe8_super_yangian.tex`.

1.4 CY-D at odd d

Correct chain-level κ_{ch} at odd d . The naive Hodge-supertrace identification $\kappa_{\text{ch}}(A_X) = \sum_q (-1)^q h^{0,q}(X)$ holds at $d = 2$ (e.g. $K3 \rightarrow 2$) but fails at $d = 3$: on $K3 \times E$ the Serre-involution cancellation forces $\kappa_{\text{ch}} = 0$ as a Hodge supertrace, while the Heisenberg-level generator count gives $\kappa_{\text{ch}}^{\text{Heis}} = 3$. The correct chain-level formula is $\kappa_{\text{ch}} = \text{str}_{F^0}(q^{L_0})$ on the Hodge-filtered piece, conjecturally reducible at $d = 3$ to a sum of $\kappa_{\text{ch}}^{\text{Heis}}$ contributions from each E -transverse Σ_2 -fibre; the statement needs a chain-level proof. See `notes/wave12_a5_cy_d_stratification.tex`.

Extension beyond McKay to non-toric compact cases. The κ_{ch} -atlas is complete for toric local surfaces (via McKay quiver) and for $K3 \times E$ (via Mukai). The compact quintic has $\kappa_{\text{ch}}^{\text{BCOV}} = -25/3$ from the one-loop BCOV weight but no first-principles $\kappa_{\text{ch}}^{\text{Heis}}$. Non-compact non-toric CY_3 (e.g. generic local $K3$) has no direct McKay input. See `notes/wave12_a5_cy_d_stratification.tex` (Open questions 1–4).

2 Frontier by universal Borcherds weight identity

2.1 Borcherds weight extensions at non-CHL orders

Borcea–Voisin threefold at $N = 5$. The Cléry–Gritsenko 8-form extension at $N \in \{5, 7, 8\}$ yields half-integer weights; a geometric CY_3 -host at these orders is conjectural. The candidate at $N = 5$ is a Borcea–Voisin threefold $Y_5 = (K3_{\mathbb{Z}_5} \times E_{\mathbb{Z}_5})/\mathbb{Z}_5$ with $(Z^{Y_5})^{-1/4} = \Phi_5$ a metaplectic fourth root of the paramodular form; the fractional-exponent Borcherds lift is a pseudo-character $S^{\text{ps}} : \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \rightarrow \mathfrak{g}_{\Phi_5}$. See `notes/wave12_a2_kappa_invariants_universal_borcherds.tex` (Open question 1) and `notes/wave12_a6_universal_borcherds_verification.tex`.

$N = 7$ order-4 gerbe Cheeger–Simons character. The Niemeier fixed-count 3 at $N = 7$ forces a singular limit in the K3 fixed-locus construction; the residual content is an order-4 discrete gerbe on the CHL orbifold whose banding cocycle is an $H^3(M_{24}, U(1))$ -valued Cheeger–Simons character. The existence of a non-singular BKM sibling at $N = 7$ is the falsifiable prediction; currently obstructed. See `notes/wave12_f5_programme_meta_audit.tex` (attack–heal 5 falsifiability, prediction (b)) and `notes/wave12_a6_universal_borcherds_verification.tex`.

$N = 8$ Mongardi–Tari–Wandel Kummer-3 fourfold. At $N = 8$ the Borcherds weight is $c_8(0)/2 = 0$, degenerate from the abelian-lattice Narain-orbit viewpoint. The candidate host is a Mongardi–Tari–Wandel Kummer-3 hyperkähler fourfold with μ_8 -gerbe banding; the fourfold breaks the CHL $\{1, 2, 3, 4, 6\}$ ladder and introduces a $d = 4$ symplectic-shift component to the Φ -image. See `notes/wave12_a6_universal_borcherds_verification.tex` (Open question 2).

2.2 Fake Monster at $d = 5$

Explicit $\text{Sp}_{K3 \times K3, E}^{\text{ch}}$ on $K3 \times K3 \times E$. The Fake Monster sibling lives at $d = 5$ (rank obstruction: the rank-24 Leech lattice does not embed into $h^{1,1}(K3 \times E)$; it does embed into a degree-(4, 20) sublattice of $\tilde{H}^*(K3 \times K3) \oplus H^1(E)$). The Φ_5^{FA} -target at $d = 5$ carries an E_5 -Poisson bracket (PTVV shift +1), and the chain-level computation on $K3 \times K3 \times E$ producing the Borcherds Fake-Monster denominator $\kappa_{\text{BKM}}(\Phi_{\text{FM}}) = 12$ is not written. See `notes/wave12_f5_programme_meta_audit.tex` (open question 4).

2.3 Niemeier and Kummer extensions

Rank- ≥ 3 lattice-polarised $\mathfrak{g}_L$ family. The Niemeier-lattice-indexed BKM superalgebras at rank ≤ 2 cover $N \in \{1, 2, 3, 4, 6\}$; the rank- ≥ 3 lattice-polarisation family (*ibid.* elliptic-surface Mordell–Weil indexing) produces a tower of lattice-polarised $\mathfrak{g}_L$ with $\kappa_{\text{BKM}}(\mathfrak{g}_L) = c_L(0)/2$ computed from the Niemeier Weyl vector. See `notes/platonic_synthesis_waves_11_through_16.tex` (Residual frontier) and `notes/wave12_a6_universal_borcherds_verification.tex`.

Elliptic-surface specialisation $(\Sigma_2, C) = (\mathcal{E}, \mathbb{P}^1)$. Specialising the (Σ_{d-1}, C) -cycle datum of the two-stage factorisation to an elliptic surface $\mathcal{E}$ with a $\mathbb{P}^1$ -base converts the six routes to $G(K3 \times E)$ into a single $\text{Aut}(K3) \times \text{SL}_2(\mathbb{Z})$ -orbit in $H_4(K3 \times E; \mathbb{Z})$ under a Mordell–Weil-indexing conjecture. See `notes/platonic_synthesis_waves_11_through_16.tex` (Residual frontier item on Mordell–Weil unification).

3 Frontier by structural conjecture

3.1 Lorgat 2020 Conjecture 1, sharpened

The output $\text{Sp}_{K3,E}^{\text{ch}}(\mathcal{F}_{K3 \times E}) \simeq U^{\text{ch}}(\mathfrak{heis}_{\text{Muk}}) \otimes U^{\text{ch}}(\mathfrak{g}_{K3}^{\text{BPS}})$ is currently identified structurally; the sharpened form promotes this to a chiral-bialgebra isomorphism on E with $\mathbf{H}_{\Delta_5}$, carrying character Δ_5^{-2} and root multiplicities equal to g_N -twisted-twined K3 elliptic genera. Five adversarial holes remain unhealed: the twisted dimension $\dim S_5(\text{Sp}_4(\mathbb{Z}), \nu_{\Delta_5}) = 1$ cite is uncorroborated; the four-derivation convergence $C_1 = C_2 = C_3 = C_4$ at $c = -214$ is normalisation-matching-open; Oberdieck–Pixton is explicit only at $N = 1$ primitive K3-classes; Cléry–Gritsenko 2013 proves automorphy, not BKM-denominator structure; and $H^3(M_{24}, U(1)) = \mathbb{Z}/12 \oplus \mathbb{Z}/2$ is computational, not theorem. Post-wave-12 status: (1, id) is theorem modulo holes 1–2; rows $N \in \{2, 3, 4, 6\}$ are conjectural. See `notes/wave12_d2_conj1_adversarial.tex` and `notes/wave12_f5_programme_meta_audit.tex` (shortest-path frontier, Vol III).

3.2 Chiral volume conjecture

The asymptotic $\lim_{N \rightarrow \infty} (1/N) \log |Z_N(X, C)| = (1/2\pi) |\text{AJ}_X(C)|$, conditional on CY-A₃, identifies the chiral-partition-function growth rate with the Abel–Jacobi period of the reference curve $C \subset X$; the physical interpretation is the on-shell action of 6d hCS on $\tilde{X} = \text{Tot}(\Omega_X^{3,0})$ with C as source. Test at small $N \in \{1, 2, 3\}$ on the quintic against the Picard–Fuchs period of Ω_X is open; large- N uniformity near conifold points (where AJ develops logarithmic singularities) is open; extension to local CY₃ replacing AJ by the Mordell integral is open. See `notes/wave12_a14_chiral_volume.tex`.

3.3 Class M E_3 bar = 6^g beyond $g = 3$

The identity $\dim H^\bullet(B_{E_3}(\mathcal{A}_\sigma); \Sigma_g) = 6^g$ is theorem at $g \leq 3$ by direct tricomplex computation, and conjectural at $g \geq 4$ pending evaluation of higher differentials $d_{r \geq 5}$ on the $g = 4$ tricomplex of support width 5. The Gevrey-1 Borel resummation of the shadow tower predicts 6^g as the summed value; the chain-level verification at $g = 4$ requires computing d_5 explicitly. An M_{24} -equivariant refinement at every g and a Kauffman-skein-module dictionary are both open. See `notes/wave12_a10_class_m_e3_bar.tex` (Open questions).

4 Frontier by physics interweave

4.1 6d hCS on compact CY_3

The one-loop counterterm for 6d hCS on a compact CY_3 is controlled by the Pontryagin class $c_2(K3 \times E) = c_2(K3)$; it is expected trivialisable by a local 6d hCS functional but no explicit Feynman-coefficient-level writing is available. The (ε, L) -regularised Costello renormalisation on the compact total space is open; the non-abelian pushforward $\text{Obs}_{\text{hCS}}^{q,E}(K3 \times E, \mathfrak{g})$ and its identification with the Lorgat GKM $\mathfrak{g}_{\Delta_5}$ as a chiral algebra is the direct physical instantiation of the sharpened Conjecture 1. See `notes/wave12_f1_6d_hcs_e3_generators.tex` (Open questions 1–2) and `notes/wave12_f3_feynman_coefficients_costello.tex` (Q4: $K3 \times E$ pushforward).

4.2 Line and surface operators

The 6d analogue of Costello–Witten 4d hCS, which furnishes the affine Yangian structure from a line operator, is not written. Line operators on 6d hCS supported on $(0,1)$ -curves transverse to generic slicing should carry Chern–Simons–Witten-type central charges; the Yangian target of 4d hCS line operators becomes a Super-Yangian target at 6d. Surface operators on $S \subset X$ should align their E_1 -shadow with the Maulik–Okounkov R -matrix of S ; transverse-direction comparison between OPE-level surface data and MO-stable-envelope cohomology is open. See `notes/wave12_f1_6d_hcs_e3_generators.tex` (items Q4–Q5).

4.3 Holographic dual

The 6d holomorphic factorisation avatar Φ_d^{FA} should sit on the asymptotic boundary of a $(d+1)$ -real-dimensional twisted M-theory in the sense of Costello–Gaiotto; the holographic pairing $(\Phi_d^{\text{FA}})_{\text{bdy}} \rightarrow Z_{\text{bulk}}$ at $d = 3$ carries the Borcherds Monster $\Phi_{10}^{1/2} = \Delta_5$ through the $(1 + 1 + 1) = 3$ sector-split of the heterotic worldsheet matter central charge $c = 24$. The $\text{AdS}_3 \times S^3 \times K3$ microstate-counting reading of $\kappa_{\text{BKM}}(\Phi_1) = 5$ as a BPS symmetry of the Strominger–Vafa black hole is conjectural; the higher- λ Gaberdiel–Gopakumar vector-model collapse $\lambda \rightarrow 1$ at large N is open. See `notes/wave12_d4_holographic_AdS3.tex` and `notes/wave12_d5_quantum_gravity_BH.tex`.

5 Frontier by cross-volume bridge

5.1 Vol I bridge

Chiral instantiation of Booth–Lazarev 2304.08409. The Booth–Lazarev curved A_∞ Quillen equivalence applies abstractly; its chiral-lane instantiation at compact CY_d with $\kappa_{\text{ch}} \neq 0$ — the curved A_∞ cotangent complex on $\overline{M}_{g,n}$ — closes Vol I B-row $K^{\kappa_{\text{ch}}} = 8$ as a cross-volume witness. See `notes/wave12_b2_cross_volume_consistency.tex` and `notes/wave12_f5_programme_meta_audit.tex`.

Chain-level MC shadow tower equal to Vol I Θ_A . The Maurer–Cartan shadow tower $\{S_k\}$ through $S_8 = 4144720/19683$ and $m_8 = 33157760/19683$, computed from the 6d hCS Feynman recursion on $\mathbb{C}^3$ at chiral-avatar level, must equal the Vol I total-shadow obstruction Θ_A on the $\text{SC}^{\text{ch,top}}$ -closed-colour component of the Φ_d -image. The genus-zero case is clean; $g \geq 1$ is Vol I conjecture MC5. See `notes/wave12_f2_bv_brst_to_chiral_ce.tex` (Q2) and `notes/wave12_a11_shadow_tower_kont.`

5.2 Vol II bridge

Chain-level $\mathcal{W}_{1+\infty}$ -endpoint of iterated Sugawara ladder. Under the curved-Dunn $H^2 = 0$ condition at $g \geq 2$, the iterated Sugawara ladder $(E_1 \rightarrow E_2 \rightarrow \cdots \rightarrow E_\infty)$ topologises to $\mathcal{W}_{1+\infty}$ at $c = 1$ (Heisenberg endpoint). The witnessing chain homotopy for the $k \rightarrow \infty$ limit is the Vol II shortest-path frontier; the three-sector Vol III reading supplies its target. See `notes/wave12_f5_programme_meta_audit.tex` (Vol II shortest-path frontier) and `notes/wave12_a11_coha_y_plus`

6 Open frontiers by single-sentence target

1. Write the chain-level factorisation-homology equation on $\text{Ran}(E)$ realising $\text{Sp}_{K3,E}^{\text{ch}}(\Phi_3^{\text{FA}}(K3 \times E)) \simeq U^{\text{ch}}(\mathfrak{heis}_{\text{Muk}}) \otimes U^{\text{ch}}(\mathfrak{g}_{K3}^{\text{BPS}})$ beyond a single reference curve (`notes/wave12_f5_programme_meta_audit.tex`, `notes/wave12_u1_two_stage_functor.tex`).
2. Prove the sharpened Lorgat 2020 Conjecture 1 chiral-bialgebra isomorphism $\text{Sp}_{K3,E}^{\text{ch}}(\Phi_3^{\text{FA}}) \cong \mathbf{H}_{\Delta_5}$ with character Δ_5^{-2} through paramodular Fourier expansion (`notes/wave12_d2_conj1_adversarial.tex`).
3. Construct explicitly the Fake Monster Φ_5^{FA} -image on $K3 \times K3 \times E$ realising the E_5 -Poisson bracket and yielding $\kappa_{\text{BKM}}(\Phi_{\text{FM}}) = 12$ (`notes/wave12_f5_programme_meta_audit.tex`).
4. Instantiate the Booth–Lazarev curved A_∞ Quillen equivalence in the chiral lane on $\overline{M}_{g,n}$ at compact CY_d with $\kappa_{\text{ch}} \neq 0$ (`notes/wave12_b2_cross_volume_consistency.tex`, `notes/wave12_f5_programme`).
5. Compute d_5 on the genus-4 tricomplex of $B_{E_3}(\mathcal{A}_\sigma)$ to upgrade class-**M** E_3 bar = 6^g from conjecture to theorem at $g \geq 4$ (`notes/wave12_a10_class_m_e3_bar.tex`).
6. State and prove the chain-level $(B_{E_1}, B_{E_2}, B_{E_\infty})$ filtered equivalence of bar complexes unifying Vol II's $\text{SC}^{\text{ch},\text{top}}$ two-coloured dioperad with Vol III's three-sector Φ_d -image stratification (`notes/wave12_f5_programme_meta_audit.tex`).
7. Construct the Borcea–Voisin threefold host Y_5 at $N = 5$ with metaplectic fourth root $(Z^{Y_5})^{-1/4} = \Phi_5$ and lift its pseudo-character $S^{\text{ps}} : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \mathfrak{g}_{\Phi_5}$ (`notes/wave12_a2_kappa_invariants_universal_bor`, `notes/wave12_a6_universal_borcherds_verification.tex`).
8. Exhibit the explicit Super–Yangian coproduct on the super-monodromy matrix $T(u)$ of $Y(\mathfrak{gl}(4|20))$ and its universal R -matrix, embedding into $\mathbf{D}_h(\mathfrak{g}_{\Delta_5})$ (`notes/wave12_a15_e8xe8_super_yangian.tex`).
9. Verify the chiral volume conjecture $\lim_{N \rightarrow \infty} (1/N) \log |Z_N(X, C)| = (1/2\pi) |\text{AJ}_X(C)|$ at $N \in \{1, 2, 3\}$ on the quintic against the Picard–Fuchs period of Ω_X (`notes/wave12_a14_chiral_volume.tex`).
10. Produce the explicit S^3 -framing trivialisation A_∞ -twisting datum on $\text{Perf}(Q^5)$ beyond the topological obstruction vanishing (`notes/wave12_a3_cy_a3_equivalence.tex`).
11. Write the $\mathcal{Z}(\text{Rep}^{E_1}(A_{K3 \times E}))$ Drinfeld-centre presentation with explicit generators, braiding matrices, and universal R -matrix (`notes/wave12_f1_6d_hcs_e3_generators.tex`).
12. Characterise the essential image of Φ_3 in Alg_{E_1} by factorisation-compatibility plus Kontsevich–Soibelman wall-crossing flatness (`notes/wave12_a3_cy_a3_equivalence.tex`).
13. State the correct chain-level κ_{ch} -formula at odd $d \geq 3$ as a sum of E -transverse Σ_2 -fibre $\kappa_{\text{ch}}^{\text{Heis}}$ contributions, superseding the naive Hodge supertrace at odd d (`notes/wave12_a5_cy_d_stratification.tex`).

14. Evaluate the Costello–Gwilliam one-loop counterterm for 6d hCS on compact $K3 \times E$ through a local $c_2(K3)$ -trivialising Chern–Simons functional (`notes/wave12_f1_6d_hcs_e3_generators.tex`, `notes/wave12_f3_feynman_coefficients_costello.tex`).
15. Compute the non-abelian pushforward $\text{Obs}_{\text{hCS}}^{\text{q},E}(K3 \times E, \mathfrak{g})$ and identify it with $\mathfrak{g}_{\Delta_5}$ as a chiral algebra on E (`notes/wave12_f1_6d_hcs_e3_generators.tex`, `notes/wave12_f3_feynman_coefficients_costello.tex`).
16. Derive the 6d analogue of the Costello–Witten affine Yangian from a line operator on 6d hCS and match it with the Super–Yangian target (`notes/wave12_f1_6d_hcs_e3_generators.tex`).
17. Align the E_1 -shadow of a surface operator $S \subset X$ in 6d hCS with the Maulik–Okounkov R -matrix of S by transverse-direction chain-level comparison (`notes/wave12_f1_6d_hcs_e3_generators.tex`).
18. Resolve the $N = 7$ order-4 gerbe Cheeger–Simons character — either exhibit a non-singular BKM sibling or confirm its obstruction through the Niemeier fixed-count 3 forcing a singular limit (`notes/wave12_a6_universal_borcherds_verification.tex`, `notes/wave12_f5_programme_meta_audio.tex`).
19. Construct the rank- ≥ 3 lattice-polarised $\mathfrak{g}_L$ family indexed by Niemeier lattices and compute its Mordell–Weil index structure under the elliptic-surface specialisation $(\Sigma_2, C) = (\mathcal{E}, \mathbb{P}^1)$ (`notes/platonic_synthesis_waves_11_through_16.tex`).
20. Bound the depth- D vanishing $P_2(D) = 0$ for the BKM Serre relation at the quantum-toroidal root-of-unity specialisation beyond $D = 3$, resolving the orbit-indexed obstruction at $D \geq 4$ (`notes/wave12_a13_bkm_serre_root_unity.tex`).